

PSFEx-Centered Inclination Analysis Report (Draft)

March 25, 2026

1 PSFEx Assessment

Metric	Value
N(FRB)	23
Mean —PSFEx Avg FWHM - ePSF Avg FWHM— [pix]	0.248
Median —PSFEx Avg FWHM - ePSF Avg FWHM— [pix]	0.232
Mean —PSFEx Moffat FWHM - ePSF Moffat FWHM— [pix]	0.118
Mean —ePSF(exp - Moffat)— [pix]	0.637
Mean —PSFEx(exp - Moffat)— [pix]	0.519
Better fit by max frac residual: ePSF count	14
Better fit by max frac residual: PSFEx count	9
Better exp-vs-Moffat agreement: ePSF count	4
Better exp-vs-Moffat agreement: PSFEx count	19

Table 1: All-FRB compact PSFEx versus ePSF comparison table.

FRB	eAvg	pAvg	dAvg	eMof	pMof	dMof	eResid%	pResid%	BestFit	BestAgr
20171020A	4.75	4.16	-0.59	3.76	3.49	-0.27	8.25	3.95	PSFEx	PSFEx
20180924B	4.74	4.57	-0.18	3.87	3.97	0.10	2.93	3.12	ePSF	PSFEx
20190102C	6.03	5.78	-0.25	5.57	5.52	-0.06	2.29	2.48	ePSF	PSFEx
20190608B	5.17	5.24	0.07	4.62	4.76	0.13	2.30	3.34	ePSF	PSFEx
20190714A	4.39	4.16	-0.23	3.53	3.56	0.03	2.66	3.52	ePSF	PSFEx
20191001A	4.54	4.38	-0.16	3.85	3.76	-0.09	2.13	3.06	ePSF	PSFEx
20200906A	4.55	4.53	-0.03	3.70	3.87	0.16	3.17	4.10	ePSF	PSFEx
20210320C	4.42	4.13	-0.29	3.52	3.49	-0.03	4.76	5.20	ePSF	PSFEx
20210410D	5.99	5.74	-0.25	5.47	5.50	0.03	3.20	2.38	PSFEx	PSFEx
20210807D	5.07	4.89	-0.18	4.39	4.30	-0.10	4.59	3.71	PSFEx	PSFEx
20211127I	5.37	5.29	-0.09	4.94	4.88	-0.06	3.27	2.66	PSFEx	PSFEx
20211203C	4.44	4.17	-0.27	3.62	3.55	-0.06	2.23	3.34	ePSF	PSFEx
20211212A	5.06	5.82	0.75	4.40	5.18	0.78	2.34	9.14	ePSF	PSFEx
20220207C	5.13	5.00	-0.13	4.38	4.45	0.07	6.05	4.41	PSFEx	PSFEx
20220307B	5.38	5.12	-0.26	4.66	4.60	-0.06	2.50	2.53	ePSF	PSFEx
20220310F	5.52	5.23	-0.29	4.92	4.80	-0.12	2.62	2.26	PSFEx	PSFEx
20220319D	4.97	4.85	-0.13	4.22	4.22	0.00	3.17	3.43	ePSF	PSFEx
20220509G	6.10	6.57	0.48	5.93	6.20	0.28	5.48	7.43	ePSF	ePSF
20220825A	5.01	4.68	-0.33	4.15	4.07	-0.08	3.39	3.01	PSFEx	PSFEx
20220912A	4.78	4.62	-0.16	4.01	3.98	-0.04	2.75	2.42	PSFEx	PSFEx
20220914A	6.81	7.08	0.27	6.72	6.74	0.02	6.75	7.39	ePSF	ePSF
20220920A	5.95	6.04	0.10	5.60	5.60	-0.00	5.88	7.84	ePSF	ePSF
20221012A	6.17	6.40	0.22	5.88	6.03	0.15	6.81	5.82	PSFEx	ePSF

2 Plots

The B -band proxy used in the multiband comparison follows:

$$B = g + 0.3130(g - r) + 0.2271.$$

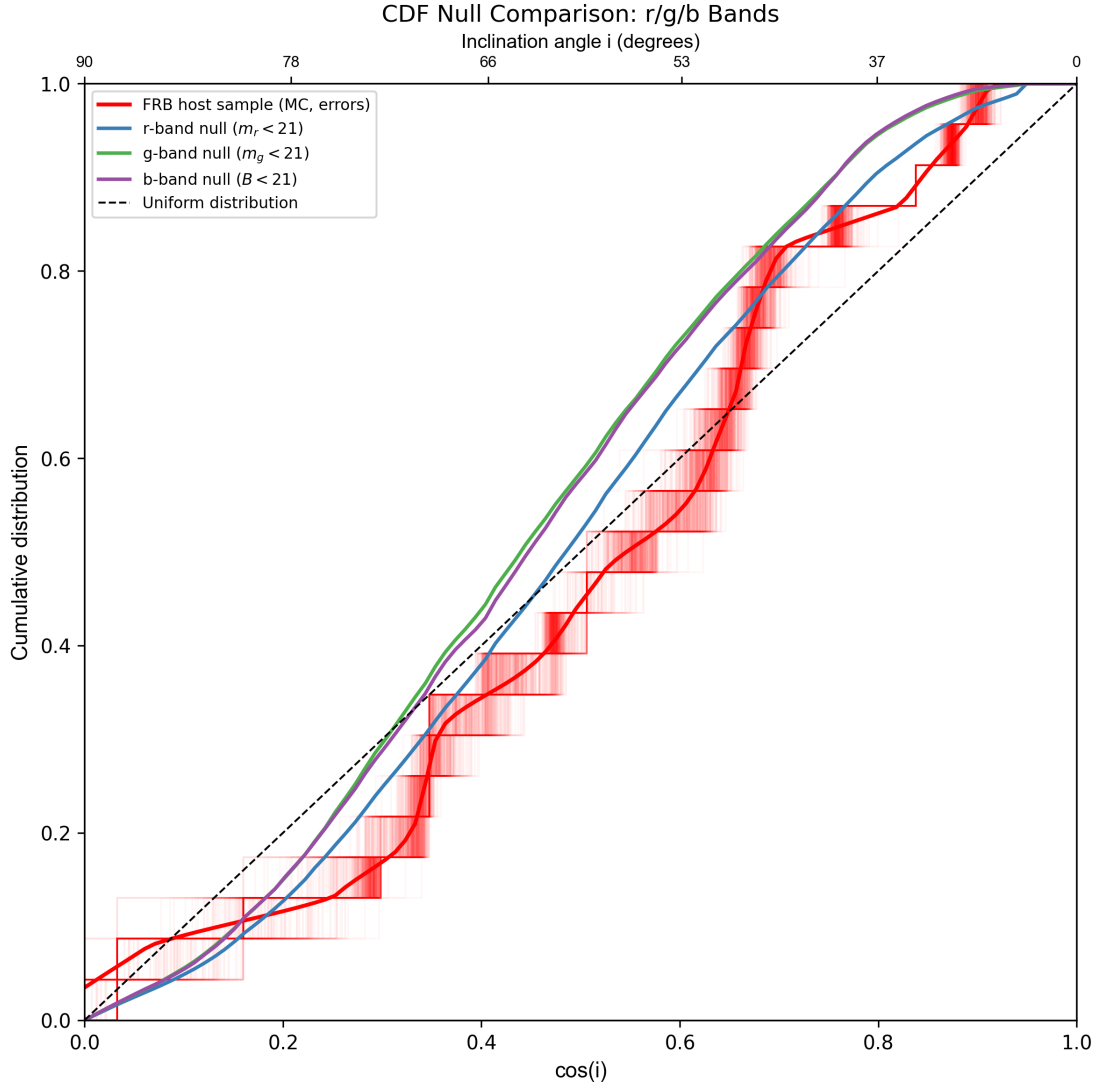


Figure 1: Multi-band CDF comparison (r/g/b nulls) with FRB host Monte Carlo envelope.

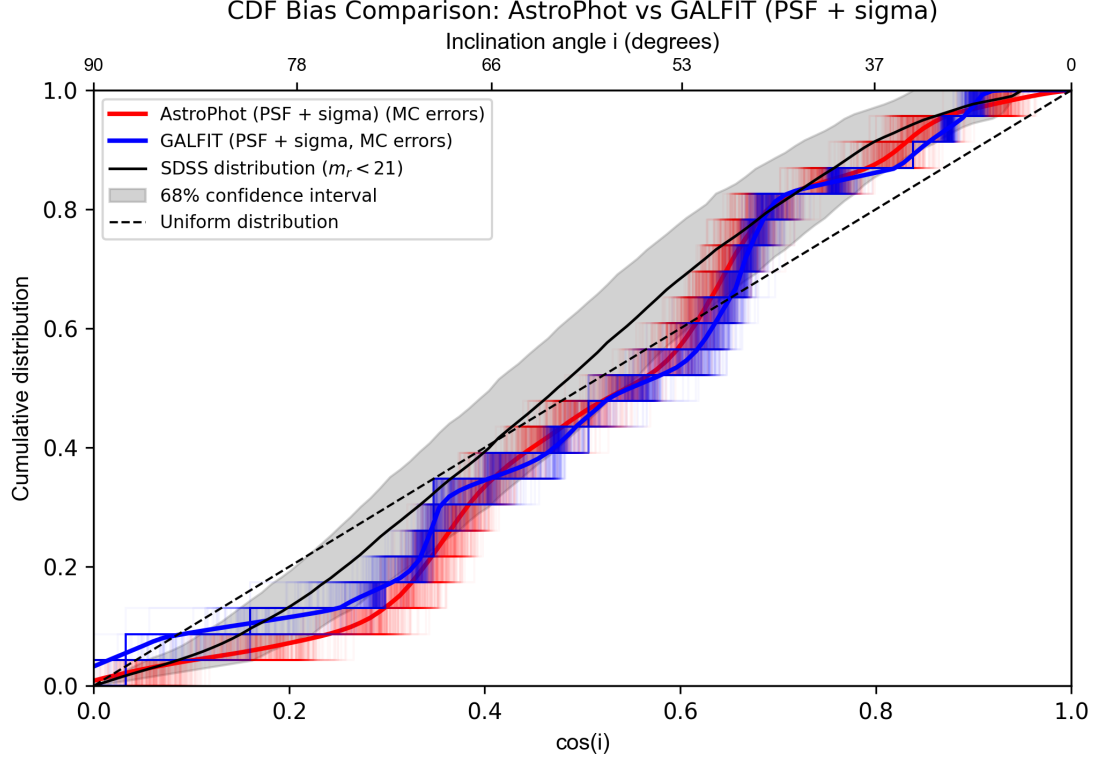


Figure 2: GALFIT versus AstroPhot comparison where both methods are shown with Monte Carlo uncertainty draws.

3 GALFIT Inclination Angles

Table 2: GALFIT PSF+sigma inclination angles with propagated uncertainties.

FRB	b/a	$\sigma_{b/a}$	i_{GALFIT} [deg]	σ_i [deg]
20171020A	0.390	0.000	70.02	0.00
20180924B	0.650	0.010	50.86	0.80
20190102C	0.400	0.030	69.30	2.18
20190608B	0.680	0.000	48.45	0.00
20190714A	0.380	0.010	70.75	0.73
20191001A	0.520	0.000	60.67	0.00
20200906A	0.320	0.000	75.23	0.00
20210320C	0.700	0.020	46.79	1.70
20210410D	0.140	0.090	90.00	6.66
20210807D	0.700	0.010	46.79	0.84
20211127I	0.910	0.010	25.03	1.46
20211203C	0.420	0.080	67.86	5.85
20211212A	0.830	0.000	34.70	0.00
20220207C	0.210	0.000	86.25	0.00
20220307B	0.650	0.030	50.86	2.41
20220310F	0.690	0.020	47.62	1.67
20220319D	0.900	0.010	26.42	1.38
20220509G	0.400	0.000	69.30	0.00
20220825A	0.630	0.050	52.43	3.98
20220912A	0.640	0.030	51.65	2.39
20220914A	0.490	0.010	62.84	0.72
20220920A	0.860	0.000	31.39	0.00

FRB	b/a	$\sigma_{b/a}$	i_{GALFIT} [deg]	σ_i [deg]
20221012A	0.550	0.000	58.47	0.00

4 Legacy Survey Inclinations vs GALFIT

Legacy/Tractor fields queried for host matching and shape-based inclination were `objid`, `ra`, `dec`, `type`, `flux_r`, `sersic`, `shape_e1`, and `shape_e2`. Legacy inclination was derived from shape terms as $|e| = \sqrt{e_1^2 + e_2^2}$, $q = (1 - |e|)/(1 + |e|)$, and $i = \cos^{-1}\left(\sqrt{(q^2 - q_0^2)/(1 - q_0^2)}\right)$ with $q_0 = 0.2$. A direct TAP schema query found `shape_e1_ivar` and `shape_e2_ivar`; we convert ivar to standard deviation with $\sigma = 1/\sqrt{\text{ivar}}$ and propagate to $\sigma_{i,\text{LS}}$ using Monte Carlo draws in (e_1, e_2) space.

Table 3: Legacy Survey inclination angles versus GALFIT inclinations, with Legacy inclination uncertainty propagated from Tractor shape ivars.

FRB	Type	n_{LS}	i_{LS}	$\sigma_{i,\text{LS}}$	i_{GF}	$\sigma_{i,\text{GF}}$	Δi
20180924B	SER	3.32	48.12	0.43	33.63	0.80	14.49
20190102C	SER	0.50	64.71	1.32	80.24	2.18	-15.53
20190608B	SER	3.49	54.32	0.11	32.52	0.00	21.80
20190714A	SER	0.53	68.70	0.38	70.75	0.73	-2.05
20191001A	SER	0.50	82.06	2.31	46.79	0.00	35.27
20200906A	SER	0.75	74.90	0.23	77.62	0.00	-2.72
20210320C	REX	–	0.00	–	45.10	1.70	-45.10
20210410D	EXP	–	74.99	1.78	90.00	6.66	-15.01
20211203C	EXP	–	62.89	4.09	56.24	5.85	6.65
20211212A	SER	1.14	35.08	0.04	32.52	0.00	2.56
20220310F	EXP	–	49.87	2.32	70.02	1.67	-20.15
20220509G	SER	2.47	59.35	0.07	62.11	0.00	-2.76
20220914A	SER	2.03	63.08	0.70	59.94	0.72	3.14
20220920A	REX	–	0.00	–	31.39	0.00	-31.39
20221012A	SER	2.27	59.53	0.34	59.21	0.00	0.32